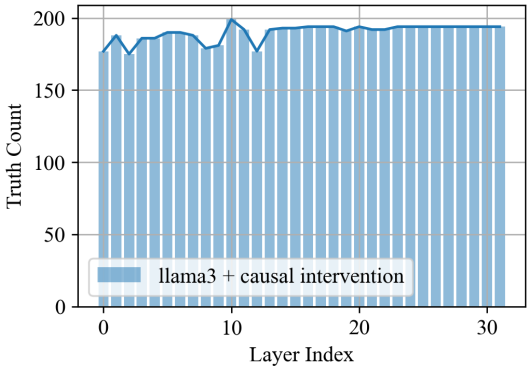
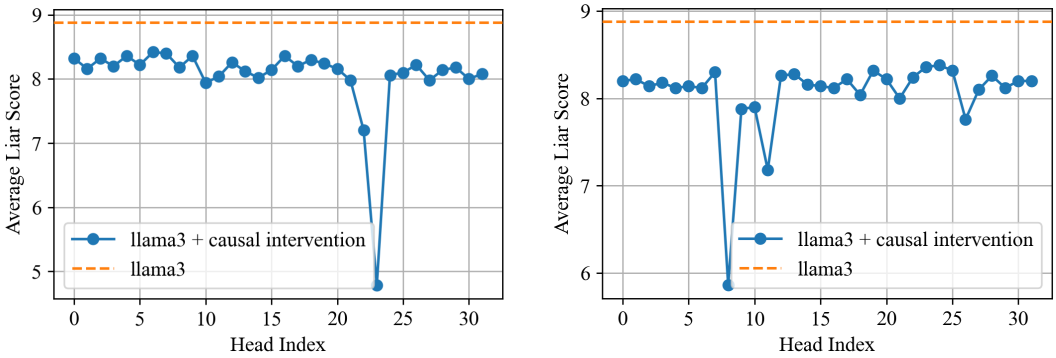


Figure 9: Zeroing out 5 layers of attention output around layer x at dummy tokens, in the truth case. In general the impact is insignificant.

concatenating outputs from the 32 heads. Thus, it is possible to do causal intervention at a finer scale, on each attention attention head separately.

We display the result of such head-wise intervention in Figure 10, only at Layers 13 and 15, which are selected based on Figure 3. It turned out that lying-related heads are very sparse. Attention heads at (Layer 13, Head 23), (Layer 15, Head 8), (Layer 15, Head 8) are found to be extraordinarily related to lying abilities, while most heads barely contribute to lying.



(a) Attention heads at Layer 13.

(b) Attention heads at Layer 15.

Figure 10: Causal intervention on each attention head, averaging over 50 prompts. We zero out attention output of the x-th attention head at Layer {13, 15}, and observe the drop in lying ability. Only very few heads are responsible for lying, which includes (Layer 13, Head 23), (Layer 15, Head 8), (Layer 15, Head 8) out of all 64 heads in these two layers. The sparse pattern shows that lying are based on very specific dedicated building blocks.

In this light, we explored the possibility of control lying simply by zeroing out the most related heads. Our proposed approach works by finding the top-k most lying-related attention heads, and zero out their outputs. Given that Llama-3.1-8B-Instruct has 32 layers, and 32 attention heads per layer, which gives $32 \times 32 = 1024$ distinctly parametrized heads in total, it is impossible to enumerate over all $\binom{1024}{k}$ combinations to select k heads. Instead, we begin by find the top-1 head that zeroing it out would harm lying ability the most, then find a second head that along with the top-1 head harmed lying the most, then find a third head that along with the top-2 heads harmed lying the most... In a word, using a greedy approach to approximate the discrete optimization for choice of attention heads.

In this setting, $P(\text{hallucination})$ and $P(\text{lying})$ are identically defined by $1 - P(\text{truth})$, where $P(\text{truth})$ is the LLM's predicted probability of the only correct token at temperature $T = 1.0$. The only difference is that $P(\text{lying})$ is computed with an extra explicit prompt that instructs the model to output wrong answers.

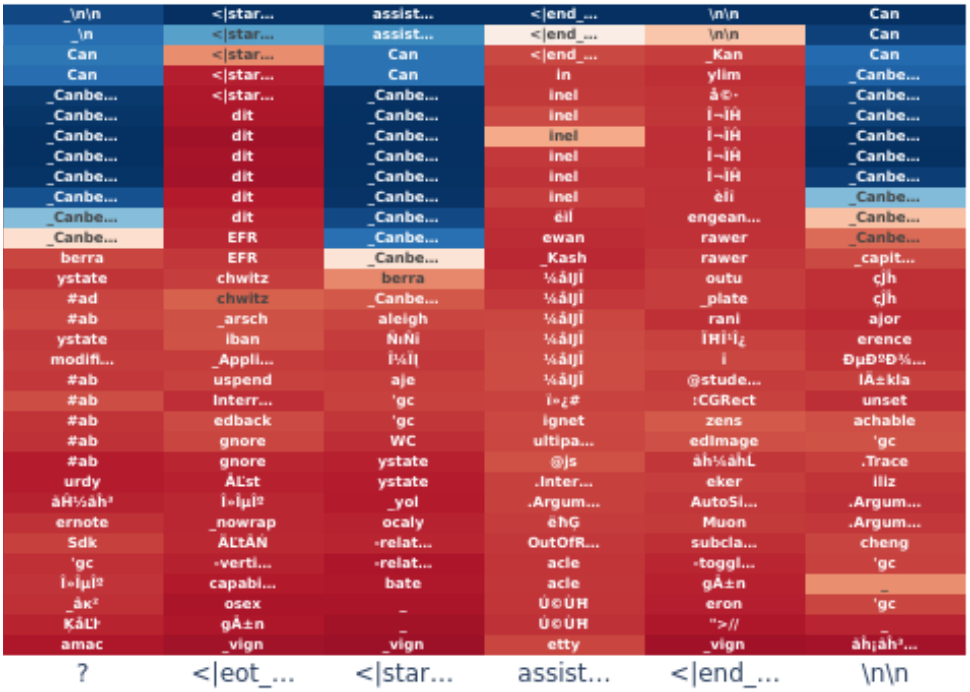


Figure 11: Truth rehearsal.

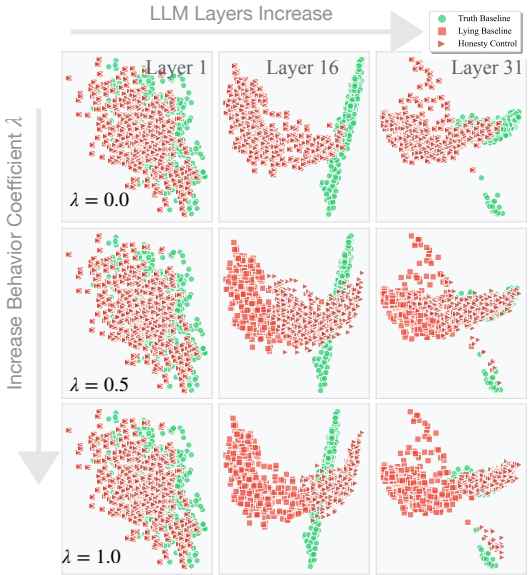
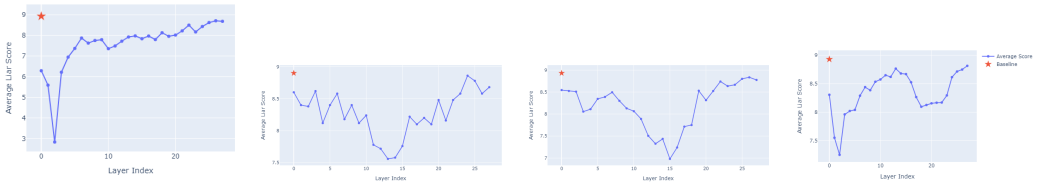


Figure 12: Principle Component Analysis.

B.3 Interpreting the Qwen model

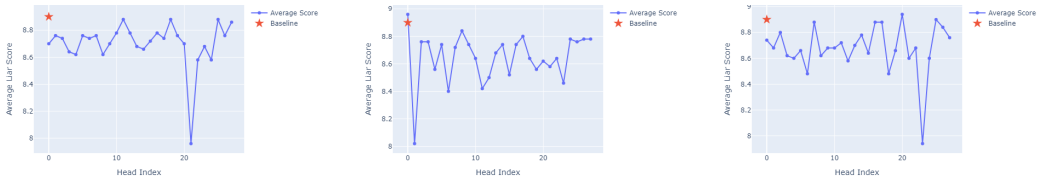
We replicated results in Figure 3 on Qwen2.5-7B-Instruct, to obtain Figure 13. We have identified similar structures in both models, despite minor differences in exact layers. In general, lying in Qwen happens at layers 13 through 23, which are later than Llama.



(a) MLP@dummies. (b) Attn@Subject→dummies. (c) Attn@Intent→dummies. (d) Attn@dummies→last.

Figure 13: Causal intervention results (averaged over 200 examples) showing the impact of zeroing out components on the liar score (lower value means the model is more truthful). The x-axis represents the center of a 5-layer window where the intervention occurs. (a) Impact of zeroing MLPs at dummy tokens. (b) Impact of blocking attention from subject tokens to dummy tokens. (c) Impact of blocking attention from intent tokens to dummy tokens. (d) Impact of blocking attention from dummy tokens to the last token. Critical layers, i.e. layers 10 through 20, for lying are highlighted by dips in the score.

We have also obtained similar results to Figure 10, as shown in Figure 14.



(a) Layer 11. (b) Layer 13. (c) Layer 15.

Figure 14: Causal intervention on each attention head, averaging over 50 prompts. We zero out attention output of the x-th attention head at Layer {11, 13, 15}, only at the last token, and observe the drop in lying ability. Only very few heads are responsible for lying, which includes (Layer 11, Head 21), (Layer 13, Head 1), (Layer 15, Head 23) out of all 84 heads (28 heads per layer) in these three layers. The sparse pattern shows that lying are based on very specific dedicated building blocks.

C Appendix: More on lie control experiments

C.1 Steering Vector Derivation and Layer Selection

The steering vectors $v_1^{\top(l)}$ representing the direction of lying at each layer l are obtained using the method described in Section 3.2. We specifically define the target behavior B as `lying` and its negation $\neg B$ as `honesty`. We constructed 200 contrastive pairs of prompts. For each pair, one prompt instructed the LLM to lie, and the other instructed it to tell the truth. For example:

- Prompt 1 (Lie): "Describe the capital of France, but intentionally state that it is Berlin."
- Prompt 2 (Truth): "Describe the capital of France factually."

We collected the hidden state activations $h_{i,\texttt{lie}}^{(l)}$ and $h_{i,\texttt{truth}}^{(l)}$ for each pair i and each layer l. Principal Component Analysis (PCA) was then performed on the difference vectors $\Delta h_i^{(l)} = h_{i,\texttt{lie}}^{(l)} - h_{i,\texttt{truth}}^{(l)}$. The first principal component, $v_1^{(l)}$, was taken as the steering vector for layer l. The decision to select layers $l \in \mathcal{L} = \{10, 11, 12, 13, 14, 15\}$ for applying steering and for calculating the mean lying signal was based on the consistency of these vectors in separating lie-related activations from truth-related ones. Specifically, these were the layers where the projection of the difference activations onto